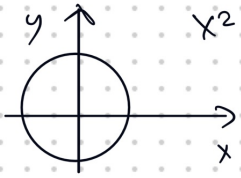
03.09.

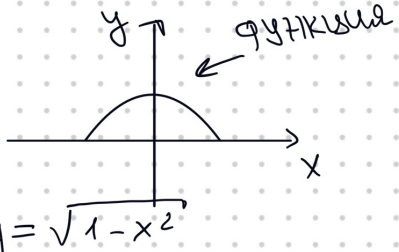
Неявные функции

$F(x,y)=0$ – уравнение

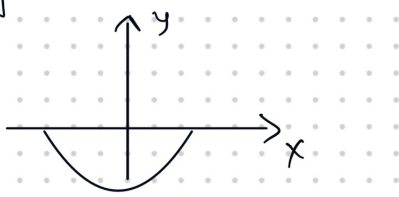


$x^2+y^2=1$

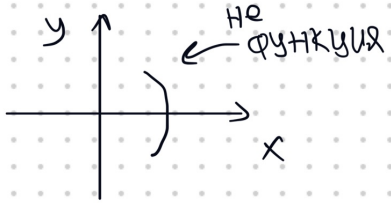
↑ не функция



$y=\sqrt{1-x^2}$



$y=-\sqrt{1-x^2}$



Теорема.

Если $f(x,y)$ непр в $U_{\delta_0}(x_0y_0)$, $F(x_0y_0)=0$ и F'_y непр в (x_0y_0) и $F'_y(x_0y_0)\neq 0$, тогда $\exists\, U_{\delta_1}(x_0)$ и $U_{\delta_2}(y_0)$

$\forall x\in U_{\delta_1}(x_0)\ \exists!\ y\in U_{\delta_2}(y_0) \rightsquigarrow F(x,y)=0$

Задана $y=f(x)$.

Следствие

Если в дополнении F'_x $\exists$ в окрестности $U_{\delta_0}(x_0y_0)$ и непр в (x_0y_0), то f диф в x_0, причем

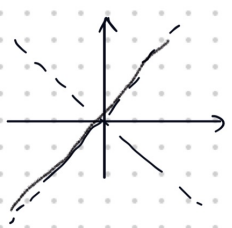
$$f'(x_0)=-\frac{F'_x(x_0y_0)}{F'_y(x_0y_0)}$$

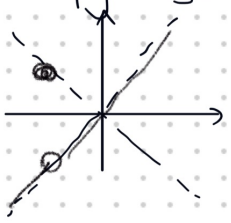
Если в дополнении F'_x и F'_y непр в $U_{\delta_0}(x_0y_0)$, то f в некоторой окрестности имеет непр производ:

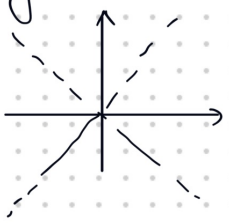
$$f'_x=-\frac{F'_x(x,f(x))}{F'_y(x,f(x))}.$$

Т1. $x^2=y^2$

а) Сколько функций $y:\mathbb{R}\to\mathbb{R}$ удовл. ур-ию?

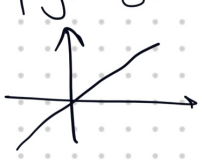


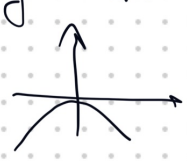




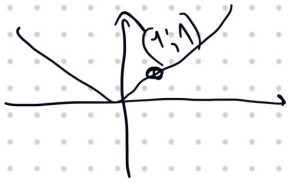
более чем счетно

б) Сколько непр функций $y:\mathbb{R}\to\mathbb{R}$ удовл. ур-ию?



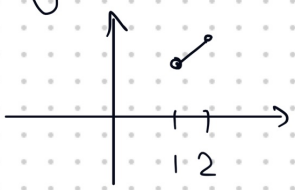


в) Сколько непр функций $y: \mathbb{R} \to \mathbb{R}$ удовл $y(1)=1$?



(2)

г) $y: [1,2] \to \mathbb{R}$ $y(1)=1$.



(1)

Теорема

$F(x,y): U_{\delta_0}(x^{(0)}, y^{(0)}) \to \mathbb{R}$; $x^\circ \in \mathbb{R}^n$, $y^\circ \in \mathbb{R}^n$.

Если $f(x,y)$ непр дчф на $U_{\delta_0}(x^\circ, y^\circ)$, $F(x^\circ, y^\circ)=0$ и $F'_y(x^\circ, y^\circ) \neq 0$, то $\exists\, U_{\delta_1}(x^\circ)$ $\exists\, U_{\delta_2}(y^\circ)$
$\forall x \in U_{\delta_1}(x^\circ)$ $\exists!\, y \in U_{\delta_2}(y^\circ) \hookrightarrow F(x,y)=0$.
т.е. на $U_{\delta_1}(x^\circ)$ задана $y=f(x)$.
При этом f непр дчф в $U_{\delta_1}(x^\circ)$ и $f'_x = -\dfrac{F'_x(x, f(x))}{F'_y(x, f(x))}$.

§3 №61.1.

$x^2 - 2y^2 + 3u^2 - yu + y = 0$ $\quad (1, 1, \frac{1}{3})$

$F(x,y,u)=0$: $u = u(x,y)$

$F'_u = 6u - y$
$F'_u = 1 \neq 0$
$F'_x = 2x$ $\quad F'_y = -4y - u + 1$ $\quad u'_x = -\dfrac{F'_x}{F'_u} = -\dfrac{2}{1} = -2$
$F'_x = 2$ $\quad F'_y = -\dfrac{10}{3}$ $\quad u'_y = -\dfrac{F'_y}{F'_u} = -\dfrac{-10/3}{1} = \dfrac{10}{3}$.

§3. 64.1. $\quad u(x,y)$ $\quad (3, -2, 2)$ $\quad x \;\; y \;\; u$

$u^3 - xu + y = 0$ $\quad F'_x = -u = -2$; $F'_u = 3u^2 - x = 9$
$F'_y = 1$

$u'_x = -\dfrac{F'_x}{F'_u} = \dfrac{2}{9}$ $\quad u'_y = -\dfrac{1}{9}$

$du = \frac{2}{9}dx - \frac{1}{9}dy.$

Система ур-ий

$F_1(x_1 \dots x_n\, y_1 \dots y_m) = 0$

$\vdots$

$F_m(x_1 \dots x_n\, y_1 \dots y_m) = 0$

$\Rightarrow \vec{F}(\vec{x}, \vec{y}) = \vec{0}$ (это не ЛП)

$\vec{x} = (x_1 \dots x_n)^T$

$\vec{y} = (y_1 \dots y_m)^T$

$\vec{F} = (F_1 \dots F_m)^T.$

Матрица Якоби:

$$\frac{\partial F}{\partial y} = \begin{pmatrix} \frac{\partial F_1}{\partial y_1} & \dots & \frac{\partial F_1}{\partial y_m} \\ \vdots & & \vdots \\ \frac{\partial F_m}{\partial y_1} & \dots & \frac{\partial F_m}{\partial y_m} \end{pmatrix}$$

Теорема

Пусть $F(\bar{x}\,\bar{y})$ непр диф в $U_{\delta_0}(x^0, y^0)$, причём $F(\bar{x}^0, \bar{y}^0) = \bar{0}$ и $\det \frac{\partial F}{\partial y}(\bar{x}^0, \bar{y}^0) \neq 0$

тогда $\exists\, U_{\delta_1}(\bar{x}^0)$ и $\exists\, U_{\delta_2}(\bar{y}^0)$:

$$\frac{\partial F}{\partial x} = \begin{pmatrix} \frac{\partial F_1}{\partial x_1} & \dots & \frac{\partial F_1}{\partial x_n} \\ \vdots & & \vdots \\ \frac{\partial F_m}{\partial x_1} & \dots & \frac{\partial F_m}{\partial x_n} \end{pmatrix}$$

$\forall \bar{x} \in U_{\delta_1}(\bar{x}^0)$ $\exists!\ \vec{y} \in U_{\delta_2}(\bar{y}^0) \hookrightarrow \bar{F}(\bar{x}\,\bar{y}) = \bar{0}$

т.е. на $U_{\delta_1}(x^0)$: $\vec{y} = \vec{f}(\bar{x})$, причём

$\bar{f}$ - непр диф и $\frac{\partial f}{\partial x} = -\left[\frac{\partial F}{\partial y}\right]^{-1}\left[\frac{\partial F}{\partial x}\right]$.

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§3 № 25 (1,2)

$u(x,y)$, $v(x,y)$

$xe^{u+v} + 2uv = 1$

$ye^{u-v} - \frac{u}{1+v} = 2x$

$u(1,2) = v(1,2) = 0.$

$(x, y, u, v) = (1, 2, 0, 0)$!

$F_1(x, y, u, v) = xe^{u+v} + 2uv - 1 = 0$

$F_2(x, y, u, v) = ye^{u-v} - \frac{u}{1+v} - 2x = 0.$

$$\frac{\partial F}{\partial \bar{y}} = \begin{pmatrix} \frac{\partial F_1}{\partial u} & \frac{\partial F_1}{\partial v} \\ \frac{\partial F_2}{\partial u} & \frac{\partial F_2}{\partial v} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & -2 \end{pmatrix}.$$

$\det = -3 \neq 0.$

$\frac{\partial F_1}{\partial u} = xe^{u+v} + 2v$ $\quad \frac{\partial F_1}{\partial u}(1,2,0,0) = 1.$

$\frac{\partial F_1}{\partial v} = xe^{u+v} + 2u$ $\quad \frac{\partial F_1}{\partial v} = 1.$

$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{\Delta}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$

$\vec{y} = \vec{f}(\vec{x})$ $\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} u(x,y) \\ v(x,y) \end{pmatrix}$

$$\frac{\partial f_2}{\partial u} = ye^{u-v} - \frac{1}{1+v} \; ; \quad \frac{\partial f_2}{\partial v}(1,2,0,0) = 1$$

$$\frac{\partial f_2}{\partial v} = -ye^{u-v} - u\frac{-1}{(1+v)^2} \cdots \frac{\partial f_2}{\partial v} = -2.$$

$$\left[\frac{\partial f}{\partial y}\right]^{-1} = -\frac{1}{3}\begin{pmatrix} -2 & -1 \\ -1 & 1 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}$$

$$g = \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} u(x,y) \\ v(x,y) \end{pmatrix}$$

$$\frac{\partial f}{\partial x} = \begin{pmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix} \qquad \frac{\partial f}{\partial x} = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{pmatrix}$$

$$\frac{\partial f_1}{\partial x} = \cdots = 1$$

$$\frac{\partial f_1}{\partial y} = 0$$

$$\frac{\partial f_2}{\partial x} = -2$$

$$\frac{\partial f_2}{\partial y} = 1.$$

$$\begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{pmatrix} = -\frac{1}{3}\begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix} = -\frac{1}{3}\begin{pmatrix} 0 & 1 \\ 3 & -1 \end{pmatrix}$$

Прошлый сем у Ира

Дифференцируемые отображения

Опр. пусть f - нек. правило, ставящее каждому $x \in G \subset \mathbb{R}^m$ в соотв $y \in \mathbb{R}^n$. Тогда f - отображение.

Обозн.: $f: G \subset \mathbb{R}^n \to \mathbb{R}^m$.

$$f \Leftrightarrow \begin{cases} y_1 = f_1(x_1 \dots x_n) \\ \vdots \\ y_m = f_m(x_1 \dots x_n) \end{cases}$$

$f_i(x_1 \dots x_n)$ - координат. функция

Опр. (дифференцируемость) $f: U_\delta(x^0) \subset \mathbb{R}^n$ $x^0 = (x_1^0 \dots x_n^0)$ говорят, что f диф в т. x^0, если $f(x) - f(x^0) = A\Delta x + o(|\Delta x|)$, $\Delta x \to 0$

$x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$ A - матрица $m \times n$.

$\Delta x = \begin{pmatrix} x_1 - x_1^0 \\ \vdots \\ x_n - x_n^0 \end{pmatrix}$ $|\Delta x| = \sqrt{\sum_{i=1}^{n} \Delta x_i^2}$.

Замечание. если f - диф в т x^0, то $A = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \dots & \frac{\partial f_1}{\partial x_n} \\ \vdots & & \vdots \\ \frac{\partial f_m}{\partial x_1} & \dots & \frac{\partial f_m}{\partial x_n} \end{pmatrix}$.

Замечание. если $m = n$, то A - квадратная матрица Якоби

$\det A$ - Якобиан

Th. пусть отображение $f: U_\delta(x^0) \subset \mathbb{R}^n \to \mathbb{R}^n$. пусть f непр. диф. в этой окрестности, $\det J|_{x_0} \neq 0$ тогда в $U_\varepsilon(y^0)$, $y^0 = f(x^0)$ опр. обратное отображение, причем оно непр. диф.

19.09.

§3 №86. $x\frac{\partial u}{\partial y} - y\frac{\partial u}{\partial x} = 0.$

$x = r\cos\varphi$
$y = r\sin\varphi$

$u = u(x,y) = \tilde{u}(r,\varphi) = \tilde{u}(r(x,y), \varphi(x,y))$

$\frac{\partial u}{\partial x} = \frac{\partial \tilde{u}}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial \tilde{u}}{\partial \varphi}\frac{\partial \varphi}{\partial x} = \frac{x}{r}\frac{\partial \tilde{u}}{\partial r} - y/r^2 \frac{\partial \tilde{u}}{\partial \varphi}$

$\frac{\partial r}{\partial x} = \frac{x}{r}$

$\frac{\partial \varphi}{\partial x} = -y/r^2$

$\frac{\partial r}{\partial y} = y/r$

$\frac{\partial \varphi}{\partial y} = x/r^2$

$\frac{\partial u}{\partial y} = \frac{\partial \tilde{u}}{\partial r}\frac{\partial r}{\partial y} + \frac{\partial \tilde{u}}{\partial \varphi}\frac{\partial \varphi}{\partial y} = y/r\frac{\partial \tilde{u}}{\partial r} + \frac{x}{r^2}\frac{\partial \tilde{u}}{\partial \varphi}$

Подставляем:

$x\left(y/r\frac{\partial \tilde{u}}{\partial r} + \frac{x}{r^2}\frac{\partial \tilde{u}}{\partial \varphi}\right) - y\left(\frac{x}{r}\frac{\partial \tilde{u}}{\partial r} - \frac{y}{r^2}\frac{\partial \tilde{u}}{\partial \varphi}\right) =$

$= \frac{xy}{r}\frac{\partial \tilde{u}}{\partial r} + \frac{x^2}{r^2}\frac{\partial \tilde{u}}{\partial \varphi} - \frac{xy}{r}\frac{\partial \tilde{u}}{\partial r} + \frac{y^2}{r^2}\frac{\partial \tilde{u}}{\partial \varphi} = 0$

$\boxed{\frac{\partial \tilde{u}}{\partial \varphi} = 0.}$ ← Функция от φ не зависит

$\tilde{u} = f(r),\ f \in C^1$

$u = f(\sqrt{x^2+y^2}),\ f \in C^1.$ ← ответ

§3 №90. Преобразовать $(y-z)\frac{\partial z}{\partial x} + (y+z)\frac{\partial z}{\partial y} = 0.$

приняв x за функцию, а $u = y - z$ за $v = y + z$ перемен.

1.) $y = \frac{v+u}{2}$

$z = \frac{v-u}{2}$

$$dy = \frac{1}{2}du + \frac{1}{2}dv$$

$$dz = \frac{1}{2}dv - \frac{1}{2}du$$

2) $x = x(u,v)$ — хотим

$$dx = \frac{\partial x}{\partial u}du + \frac{\partial x}{\partial v}dv$$

3) $z = z(x,y)$ — есть

$$dz = \frac{\partial z}{\partial x}dx + \frac{\partial z}{\partial y}dy$$

4) 2 подставляем в 3:

$$dz = \frac{\partial z}{\partial x}\left(\frac{\partial x}{\partial u}du + \frac{\partial x}{\partial v}dv\right) + \frac{\partial z}{\partial y}\left(\frac{1}{2}du + \frac{1}{2}dv\right) =$$

$$= \left(\frac{\partial z}{\partial x}\frac{\partial x}{\partial u} + \frac{1}{2}\frac{\partial z}{\partial y}\right)du + \left(\frac{\partial z}{\partial x}\frac{\partial x}{\partial v} + \frac{1}{2}\frac{\partial z}{\partial y}\right)dv.$$

5) Сравниваем 4 и 1: пр-ло Крамера

$$\begin{cases} \frac{\partial z}{\partial x}\frac{\partial x}{\partial u} + \frac{1}{2}\frac{\partial z}{\partial y} = -\frac{1}{2} \\ \frac{\partial z}{\partial x}\frac{\partial x}{\partial v} + \frac{1}{2}\frac{\partial z}{\partial y} = \frac{1}{2} \end{cases}$$

$$\frac{\partial z}{\partial x} = \frac{1}{\frac{\partial x}{\partial v} - \frac{\partial x}{\partial u}}$$

$$\frac{\partial z}{\partial y} = \frac{\frac{\partial x}{\partial u} + \frac{\partial x}{\partial v}}{\frac{\partial x}{\partial v} - \frac{\partial x}{\partial u}}$$

6) 5) в ур-ие:

$$u\left(\frac{\partial x}{\partial v} - \frac{\partial x}{\partial u}\right)^{-1} - v\left(\frac{\partial x}{\partial v} - \frac{\partial x}{\partial u}\right)^{-1}$$

$$\left(\frac{\partial x}{\partial u} + \frac{\partial x}{\partial v}\right) = 0$$

$$\boxed{v\frac{\partial x}{\partial u} + v\frac{\partial x}{\partial v} = u}$$

← искомое

№51.2

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\partial y} + y^2\frac{\partial^2 u}{\partial y^2} = 0.$$

$\frac{\partial r}{\partial x} = x/r$ $\quad \frac{\partial \varphi}{\partial x} = -y/r^2$

$\frac{\partial r}{\partial y} = y/r$ $\quad \frac{\partial \varphi}{\partial y} = x/r^2$

$x = r\cos\varphi$

$y = r\sin\varphi$

$r = \sqrt{x^2+y^2}$

$u = u(x,y) = \tilde{u}(r,\varphi) = \tilde{u}(r(x,y), \varphi(x,y))$

$$\frac{\partial u}{\partial x} = \frac{\partial \tilde{u}}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial \tilde{u}}{\partial \varphi}\frac{\partial \varphi}{\partial x} = \frac{x}{r}\frac{\partial \tilde{u}}{\partial r} - \frac{y}{r^2}\frac{\partial \tilde{u}}{\partial \varphi}.$$

$$\frac{\partial^2 u}{\partial x^2} = \left(\frac{1}{r} - \frac{x^2}{r^3}\right)\frac{\partial \tilde{u}}{\partial r} + \frac{x}{r}\left(\frac{\partial^2 \tilde{u}}{\partial r^2}\frac{x}{r} - \frac{\partial^2 \tilde{u}}{\partial r\partial\varphi}\frac{y}{r^2}\right) +$$

$$+\frac{2yx}{r^4}\frac{\partial \tilde{u}}{\partial \varphi} - \frac{y}{r^2}\left(\frac{\partial^2 \tilde{u}}{\partial r\partial\varphi}\frac{x}{r} + \frac{\partial^2 \tilde{u}}{\partial \varphi^2}\left(-\frac{y}{r^2}\right)\right)$$

$$\frac{\partial^2 u}{\partial x\partial y} = \frac{x}{-r^2}\frac{y}{r}\frac{\partial \tilde{u}}{\partial r} + \frac{x}{r}\left(\frac{\partial^2 \tilde{u}}{\partial r^2}\frac{y}{r} + \frac{\partial^2 \tilde{u}}{\partial r\partial\varphi}\frac{x}{r^2}\right) - \left(\frac{1}{r^2} - \frac{2y}{r^3}\frac{y}{r}\right)\frac{\partial \tilde{u}}{\partial \varphi}$$

$$- \frac{y}{r^2}\left(\frac{\partial^2 \tilde{u}}{\partial \varphi}\frac{\partial \tilde{u}}{\partial r}\frac{y}{r} + \frac{\partial^2 \tilde{u}}{\partial \varphi^2}\frac{x}{r^2}\right).$$

$$\frac{\partial u}{\partial y} = \frac{\partial \tilde{u}}{\partial r}\frac{\partial r}{\partial \varphi} + \frac{\partial \tilde{u}}{\partial \varphi}\frac{\partial \varphi}{\partial y} = \frac{y}{r}\frac{\partial \tilde{u}}{\partial r} + \frac{y}{r^2}\frac{\partial \tilde{u}}{\partial \varphi}$$

$$\frac{\partial^2 u}{\partial y^2} = \left(\frac{1}{r} - \frac{y}{r^2}\frac{y}{r}\right)\frac{\partial \tilde{u}}{\partial r} + \frac{y}{r}\left(\frac{\partial^2 \tilde{u}}{\partial r}\frac{y}{r} + \frac{\partial^2 u}{\partial r\partial\varphi}\frac{x}{r^2}\right) -$$

$$- \frac{\partial x}{r^3}\frac{y}{r}\frac{\partial \tilde{u}}{\partial \varphi} + \frac{x}{r^2}\left(\frac{\partial^2 \tilde{u}}{\partial r\partial\varphi}\frac{y}{r} + \frac{\partial^2 \tilde{u}}{\partial \varphi^2}\frac{x}{r^2}\right).$$

Подставляем:

$$x^2\left(\left(\frac{1}{r} - \frac{x^2}{r^3}\right)\frac{\partial \tilde{u}}{\partial r} + \frac{x}{r}\left(\frac{\partial^2 \tilde{u}}{\partial r^2}\frac{x}{r} - \frac{\partial^2 \tilde{u}}{\partial r\partial\varphi}\frac{y}{r^2}\right) + \frac{2yx}{r^4}\frac{\partial \tilde{u}}{\partial \varphi} -\right.$$

$$- \frac{y}{r^2}\left(\frac{\partial^2 \tilde{u}}{\partial r\partial\varphi}\frac{x}{r} - \frac{\partial^2 \tilde{u}}{\partial \varphi^2}\frac{y}{r^2}\right) + 2xy\left(-\frac{xy}{r^3}\frac{\partial \tilde{u}}{\partial r} + \frac{x}{r}\left(\frac{\partial^2 \tilde{u}}{\partial r^2}\frac{y}{r} + \right.\right.$$

$$\left.\left. + \frac{x}{r^2}\frac{\partial^2 \tilde{u}}{\partial r\partial\varphi}\right) - \left(\frac{1}{r^2} - \frac{2y^3}{r^4}\right)\frac{\partial \tilde{u}}{\partial \varphi} - \frac{y}{r^2}\left(\frac{y}{r}\frac{\partial^2 \tilde{u}}{\partial \varphi\partial r} + \frac{x}{r^2}\frac{\partial^2 \tilde{u}}{\partial \varphi^2}\right) +\right.$$

$$+y^2\left(\left(\frac{1}{r}-\frac{y}{r^2}\frac{y}{r}\right)\frac{\partial\tilde{u}}{\partial r}+\frac{\tilde{u}}{r}\left(\frac{\partial^2\tilde{u}}{\partial r^2}\frac{y}{r}+\frac{\partial^2\tilde{u}}{\partial r\partial\varphi}\frac{x}{r^2}\right)-\frac{2x}{r^3}\frac{y}{r}\frac{\partial\tilde{u}}{\partial\varphi}+\right.$$

$$\left.+\frac{x}{r^2}\left(\frac{\partial^2\tilde{u}}{\partial r\partial\varphi}\frac{y}{r}+\frac{\partial^2\tilde{u}}{\partial\varphi^2}\frac{x}{r^2}\right)\right)=0.$$

$$\frac{\partial^2\tilde{u}}{\partial r}\Big|$$

26.09.

Условный экстремум

Пусть на области $G\subset\mathbb{R}^n$ заданы $f(x), \varphi_1(x), \dots \varphi_m(x)$. $m<n$. Пусть $E=\{x\in\mathbb{R}^n \mid \varphi_1(x)=\dots=\varphi_m(x)=0\}$.

Опр Точка $x^0\in G$ наз (строгим) условием максимума функции f при условиях $\varphi_1(x)=0 \dots \varphi_m(x)=0$, если $\exists\ \delta>0$: $\forall x\in \mathring{U}_\delta(x^0)\cap E$

↳ $f(x)\ (<)\le f(x^0)$.

Необходимое условие условного экстремума

Пусть $f(x), \varphi_1(x), \dots \varphi_m(x)$ непр диф в $U(x^0)$ и при этом $\operatorname{rank}\left(\frac{\partial\varphi_i}{\partial x_j}\right)(x^0)=m$.

Если точка x^0 яв условным экстремумом, то

$\frac{\partial L}{\partial x_i}(x^0)=0$;

$$L=f(x)+\sum_{i=1}^{m}\lambda_i\varphi_i(x)$$

↑ функция Лагранжа $\lambda_i\in\mathbb{R}$.

Достаточное условие:

Пусть $f\ \varphi_1 \dots \varphi_m$ дважды диф в $U(x^0)$ и выполнено необх. условие. Если при равенствах

$$d\varphi_i(x^0)=\sum_{j=1}^{n}\frac{\partial\varphi_i}{\partial x_j}(x^0)\,dx_j=0,$$

матрица Гессе.

$i = 1 \dots m$ квадратичная форма d^2L является:

а) "+" определена $\Rightarrow x^0$ - строгий услов. min.

б) "−") определена $\Rightarrow x^0$ - строгий услов max.

в) не явл. знакоопр. $\Rightarrow x^0$ - не усл. экстр.

§5 № 21(2)

$u = 1 - 4x - 8y$; $x^2 - 8y^2 = 8$

$x^2 - 8y^2 - 8 = 0.$

$L = 1 - 4x - 8y + \lambda(x^2 - 8y^2 - 8).$

Необх. условия:

$$\begin{cases} \frac{\partial L}{\partial x} = -4 + 2\lambda x = 0. \\ \frac{\partial L}{\partial y} = -8 - 16\lambda y = 0. \\ x^2 - 8y^2 - 8 = 0 \end{cases}$$

$\lambda \neq 0.$

$x = \frac{2}{\lambda}$; $y = -\frac{1}{2\lambda}.$

$\frac{4}{\lambda^2} - \frac{8}{4\lambda^2} - 8 = 0.$

$\frac{2}{\lambda^2} = 8 \quad \hookrightarrow \lambda = \pm\frac{1}{2}.$

1) $\lambda = \frac{1}{2}$ $(4, -1)$

2) $\lambda = -\frac{1}{2}$ $(-4, 1)$ ← кандидаты.

Достаточное условия.

Второй диф: $\frac{\partial^2 L}{\partial x^2} = 2\lambda$

$\frac{\partial^2 L}{\partial y^2} = -16\lambda$

$\frac{\partial^2 L}{\partial x \partial y} = 0.$

матрица Гессе

$d^2L = (dx\,dy)\,H \begin{pmatrix} dx \\ dy \end{pmatrix} = (dx\,dy) \begin{pmatrix} 2\lambda & 0 \\ 0 & -16\lambda \end{pmatrix} \begin{pmatrix} dx \\ dy \end{pmatrix} = 2\lambda dx^2 - 16\lambda dy^2$

$= 2\lambda(dx^2 - 8dy^2)$

$= 2\lambda\left(\frac{64y^2}{x^2} - 8\right)dy^2 =$

$dy = 0.$

$\varphi = x^2 - 8y^2 - 8$

$2x\,dx - 16y\,dy = 0.$

$x\,dx = 8y\,dy$. Подставим $\Big| = 2\lambda\left(64\frac{y^2}{x^2} - 8\right)dy^2$.

$dx = \frac{8y}{x}dy$

$\boxed{\lambda = -7}$

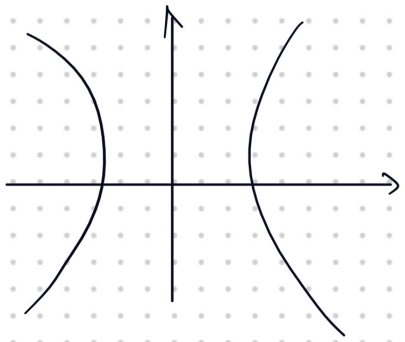
$d^2L\big|_{(1)} = 2\cdot\frac{1}{2}\left(64\,\frac{1}{16} - 8\right)dy^2 = -4dy^2 < 0$. $\hookrightarrow (4;-1)$ - max.

$d^2L\big|_{(2)} = 2\left(-\frac{1}{2}\right)\left(64\,\frac{1}{16} - 8\right)dy^2 = 4dy^2 > 0 \hookrightarrow (-4;1)$ - min.

$\boxed{u = 9}$

Второй способ:

$x^2 - 8y^2 = 8$; $u = 1 - 4x - 8y$.



Параметризуем:

гипербола.

1) $x = 2\sqrt{2}\,\mathrm{ch}\,t$; $t \in \mathbb{R}$.
$y = \mathrm{sh}\,t$
$u = 1 - 8\sqrt{2}\,\mathrm{ch}\,t - 8\,\mathrm{sh}\,t$.

$0 = u'_t = -8\sqrt{2}\,\mathrm{sh}\,t - 8\,\mathrm{ch}\,t = 0$

экстремум функции одной переменной.

2) $x = -2\sqrt{2}\,\mathrm{ch}\,t$; $y = \mathrm{sh}\,t$
$u = 1 + 8\sqrt{2}\,\mathrm{ch}\,t - \mathrm{sh}\,t$.

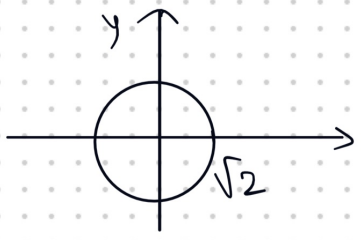
$u'_t = 0$.

§5 № 26.(3).

$u = xy + yz$; $x^2 + y^2 = 2$
$y + z = 2$
$y > 0$.

$z = 2 - y$.

$u = xy + 2y - y^2$; $x^2 + y^2 = 2$
$y > 0$.



$x = \sqrt{2}\cos\varphi$
$y = \sqrt{2}\sin\varphi$.
$\varphi \in (0;\pi)$;

$$u = 2\cos\varphi\sin\varphi + 2\sqrt{2}\sin\varphi - 2\sin^2\varphi.$$

$$u'_\varphi = 2\cos^2\varphi - 2\sin^2\varphi + 2\sqrt{2}\cos\varphi - 4\sin\varphi(\cos\varphi) = 0.$$

$$\cos^2\varphi - \sin^2\varphi + \sqrt{2}\cos\varphi - \sin 2\varphi = 0.$$

$$2\cos^2\varphi - 1 + \sqrt{2}\cos\varphi - 2\sin\varphi\cos\varphi = 0.$$

$$\sin\varphi = \frac{2\cos^2\varphi - 1 + \sqrt{2}\cos\varphi}{2\cos\varphi} = 2\cos\varphi - \frac{1}{2\cos\varphi} + \frac{\sqrt{2}}{2}.$$

$$\sin^2\varphi = 4\cos^2\varphi + \frac{1}{4}\cos^2\varphi - 2\cdot\frac{2}{2} + \frac{2}{4} + 2\cdot\frac{\sqrt{2}}{2}\left(2\cos\varphi - \frac{1}{2}\cos\varphi\right)$$

$$= 4\cos^2\varphi + \frac{1}{4\cos^2\varphi} - 2 + \frac{1}{2} + 2\sqrt{2}\cos\varphi - \frac{\sqrt{2}}{2}\cos\varphi =$$

$$= 4\cos^2\varphi + \frac{1}{4\cos^2\varphi} - \frac{3}{2} + \frac{3}{2}\sqrt{2}\cos\varphi.$$

$$\sin^2\varphi + \cos^2\varphi = 1.$$

$$5\cos^2\varphi + \frac{1}{4\cos^2\varphi} - \frac{3}{2} + \frac{3}{2}\sqrt{2}\cos\varphi = 1.$$

$$5\cos^2\varphi + \frac{1}{4\cos^2\varphi} - \frac{5}{2} + \frac{3}{2}\sqrt{2}\cos\varphi = 0. \quad | \cdot 4\cos^2\varphi$$

$$20\cos^4\varphi + 1 - 10\cos^2\varphi + 6\sqrt{2}\cos^3\varphi = 0. \qquad t = 2\sqrt{2}.$$

$$20\cos^4\varphi + 6\sqrt{2}\cos^3\varphi - 10\cos^2\varphi + 1 = 0. \qquad z = \pm 1$$

$$20\frac{z^4}{2} + 6\sqrt{2}\frac{z^3}{4\sqrt{2}} - 10\frac{z^2}{2} + 1 = 0.$$

$$10z^4 + \frac{3}{2}z^3 - 5z^2 + 1 = 0.$$

$$\cos\varphi = \pm\frac{1}{\sqrt{2}}$$

$$\cos\varphi = \frac{-1 \pm \sqrt{3}}{2\sqrt{2}}$$

① $(x, y) = (1, 1)$

② $(x, y) = (-1, 1)$

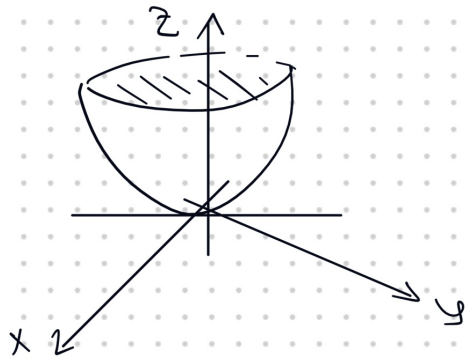
③ $(x, y) = \left(\frac{-1-\sqrt{3}}{2}; \frac{\sqrt{3}-1}{2}\right)$.

ТЮТЮМ.

№ 31.8.

$u = x + y + z$; $x^2 + y^2 \le z \le 1$.

Найти наиб и наим. знач.



1. Проверим экстремума внутри

$\nabla u = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ нет экстремумов внутри области:

$(\nabla u \neq 0)$.

2. Крышечка

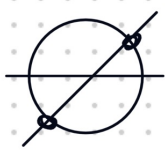
$z = 1$: $x^2 + y^2 \le 1$

$u = x + y + 1$; $x^2 + y^2 \le 1$

• проверяем внутри круга

$\nabla u = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

• на границе круга:



$u = x + y + 1$

$x^2 + y^2 = 1$

$x = \cos\varphi$

$y = \sin\varphi$.

$u = \cos\varphi + \sin\varphi + 1$

$u' = -\sin\varphi + \cos\varphi = 0$.

$\cos\varphi = \sin\varphi$.

$\varphi = \frac{\pi}{4} + \pi k$.

$x = \frac{1}{\sqrt{2}}$; $y = \frac{1}{\sqrt{2}}$; $u = 1 + \sqrt{2}$. ← экстремум

$x = -\frac{1}{\sqrt{2}}$ $y = -\frac{1}{\sqrt{2}}$ $u = 1 - \sqrt{2}$ ← экстремум.

3. Тарелочка

$x^2 + y^2 = z$; $u = x + y + z$

$x^2 + y^2 \le 1$

открытая:

$x^2 + y^2 = z$

$x^2 + y^2 < 1$.

граница

$x^2 + y^2 = z$.

$x^2 + y^2 = 1$

← "граница круга".

$u = x + y + x^2 + y^2$

$x^2 + y^2 \le 1$

$\nabla u = \begin{pmatrix} 1+2x \\ 1+2y \end{pmatrix} = 0$. $\quad x = -\frac{1}{2}$ ← экстремум.

$y = -\frac{1}{2}$.

$u = -\frac{1}{1} + \frac{1}{2} = -\frac{1}{2}$.

Ответ: $M = 1 + \sqrt{2}$

$m = -1/2$.

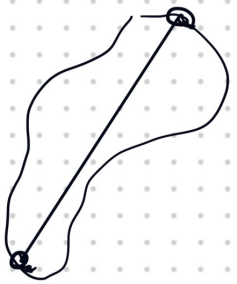
03.10. Кратные интегралы

I. Определение

Пусть X - измеримое мн-во на $\mathbb{R}^n$.

$\tau(X) = \{X_j\}_{j=1}^N : \mu(X_j \cap X_k) = 0 \; ; \; j \neq k$

$X = \bigcup_{j=1}^{N} X_j$.



$\operatorname{diam} Y = \sup_{y_1, y_2 \in Y} \rho(y_1, y_2) \; ; \; Y \in \mathbb{R}^n$

↖ диаметр

$|\tau| = \max_{j=1\ldots N} \operatorname{diam} X_j$

↖ мелкость разбиения

В каждом X_j выберем ξ_j

$\{\xi\}_{j=1}^N$ - набор точек.

Пусть f определена на $X \subset \mathbb{R}^n$. Говорят, что f интегрир на X по Риману, если $\exists\, I \in \mathbb{R}$.

$\forall \varepsilon > 0 \quad \exists \delta > 0$

$\forall \tau(X), \; |\tau| < \delta \quad \forall \{\xi\}_{j=1}^N$

↳ $|\sigma(f, \tau, \{\xi\}) - I| < \varepsilon$.

$\sigma = \sum_{j=1}^{N} f(\xi_j) \mu(X_j)$. ← сумма Римана

I. Кратный интеграл. Обозначения:

$$\int_X f(x)\,dx = \int_X \dots \int f(x)\,dx_1 \dots dx_n.$$

II. Критерий Коши

Пусть f определена на измеримом $X \subset \mathbb{R}^n$

f интегрир $\Leftrightarrow \forall \varepsilon > 0\ \exists \delta > 0$: $\forall \tau',\ |\tau'| < \delta\ \forall \{\xi'_j\}_{j=1}^{N'}$, $\forall \tau'',\ |\tau''| < \delta\ \forall \{\xi''_j\}_{j=1}^{N''}$

↳ $|\sigma(f, \tau', \{\xi'_j\}) - \sigma(f, \tau'', \{\xi''_j\})| < \varepsilon.$

Теорема.

Если функция интегрир. на открытом м-ве то она огр. на нем.

III Критерий Дарбу.

Пусть f огр на измеримом $X \subset \mathbb{R}^n$.

$$S_\tau = \sum_{j=1}^{N} m_j \mu(X_j);\ m_j = \inf_{X_j} f(x)$$

$$S_\tau = \sum_{j=1}^{N} M_j \mu(X_j);\ M_j = \sup_{X_j} f(x).$$

$$I^* = \inf_\tau S_\tau$$

$$I_* = \sup_\tau S_\tau$$

Теорема.

Пусть f – огр на измеримом X, тогда

1. f интегрир $\Leftrightarrow I^* = I_*$.
2. $\forall \varepsilon > 0\ \exists \tau$: $S_\tau - s_\tau < \varepsilon$.

IV Достаточные условия интегрир.

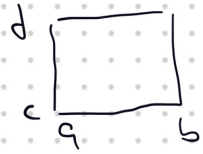
Пусть f огр. на замкнутом, измеримом $X \subset \mathbb{R}^n$. и м-во точек разрыва функции

f имеет меру 0. Тогда f - интегр на X.

§8 № 36.1.

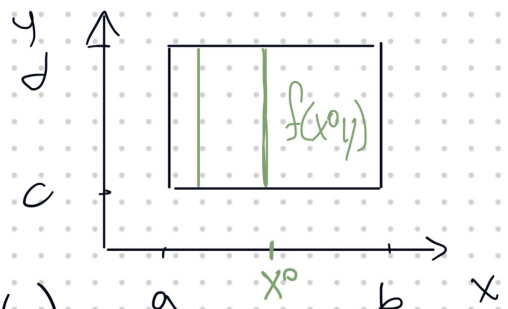
Указать измеримое X и определенную на нем f такие, что $|f|$ - интегр, а f - не интегр.

$$f(x) = \begin{cases} -1, & x \in [0;1] \cap \mathbb{Q} \\ 1, & x \in [0;1] \cap (\mathbb{R} \setminus \mathbb{Q}) \end{cases}$$



Т6. Пусть $f(x,y)$ определена на $[a,b] \times [c,d]$, $a<b$, $c<d$

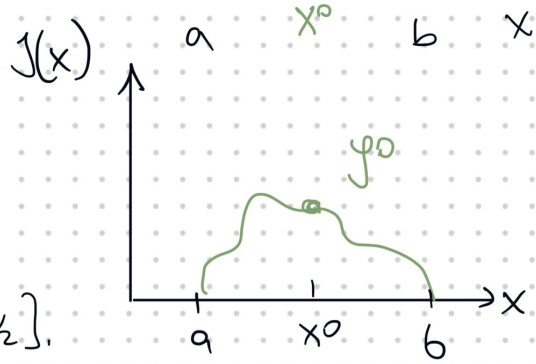
а) Верно ли, что если $f(x_0,y)$ интегр на $[c,d]$ $\forall x_0$ и $J(x) = \int_a^b f(x,y)\,dy$ - интегр, то кратный интегр. $\exists$: $\int_a^b \int_c^d f(x,y)\,dx\,dy$.



с двойной

$\int_a^b \left(\int_c^d f(x,y)\,dy \right) dx$ - существ.

(повторный интеграл)



Нет :(

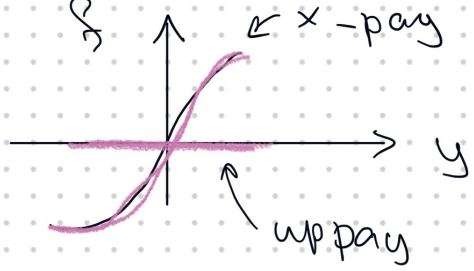
$[a,b] = [0;1]$ $[c,d] = [-\pi/2; \pi/2]$.

$$f(x,y) = \begin{cases} \sin y, & x \in [0,1] \cap \mathbb{Q} \\ 0, & x \in [0,1] \cap (\mathbb{R} \setminus \mathbb{Q}) \end{cases}$$

$$\int_c^d f(x,y)\,dy = \int_{-\pi/2}^{\pi/2} f(x,y)\,dy \quad \begin{matrix} x \in \mathbb{Q} \to \int_{-\pi/2}^{\pi/2} \sin y\,dy = 0 \\ x \in \mathbb{R} \setminus \mathbb{Q} \to \int_{-\pi/2}^{\pi/2} 0\,dy = 0. \end{matrix}$$

$$\int_a^b \left(\int_c^d f(x,y)\,dy \right) dx = \int_c^b 0\,dx = 0.$$

$\int_0^1 \int_{-\pi/2}^{\pi/2} f(x,y)\,dx\,dy$ - не существует.



$I^* > 0$

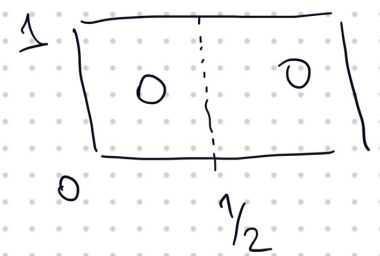
$I_* < 0$

б) Верно ли, что если f интегр на $[a,b]\times[c,d]$ то $f(x_0,y)$ интегр на $[c,d]$ $\forall x_0 \in [a,b]$.

$[a,b] = [0,1]$

$[c,d] = [0,1]$.

$$f(x,y) = \begin{cases} 0, & x \neq 1/2 \\ 1, & x = 1/2,\ y \in [0,1]\cap\mathbb{Q} \\ 0, & x = 1/2,\ y \in [0,1]\cap(\mathbb{R}\setminus\mathbb{Q}). \end{cases}$$



мера мн-ва точек разрыва 0. $\Rightarrow$

$\int_a^b \int_c^d f(x,y)\,dx\,dy \in \mathbb{R}$

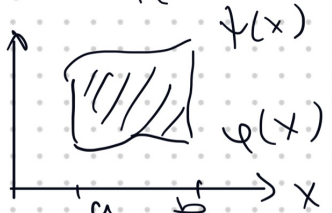
$\int_0^1 f(x_0,y)\,dy$, $x_0 = 1/2$ - не существует.

V. Сведение кратного интеграла к повторному

Опр. Мн-во $X \subset \mathbb{R}^2$ наз. элементарным относ. Оу, если

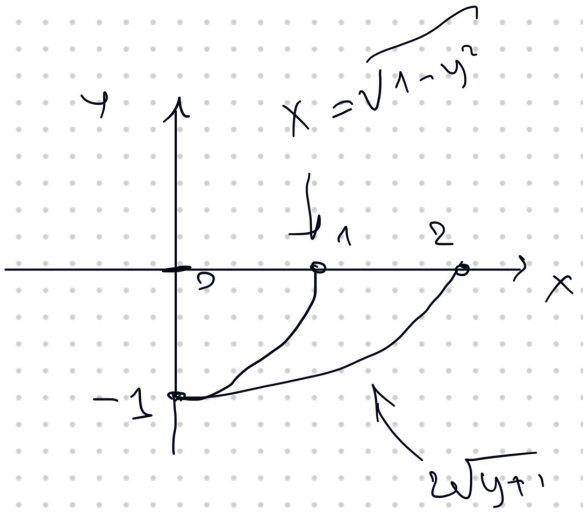
$X = \{(x,y) \in \mathbb{R}^2 \mid a \le x \le b,\ \varphi(x) \le y \le \psi(x)\}$, где

$\varphi(x)$ и $\psi(x)$ непр. на $[a,b]$



Теорема. Пусть f непр. на элементарной относ. Оу мн-ве X. Тогда

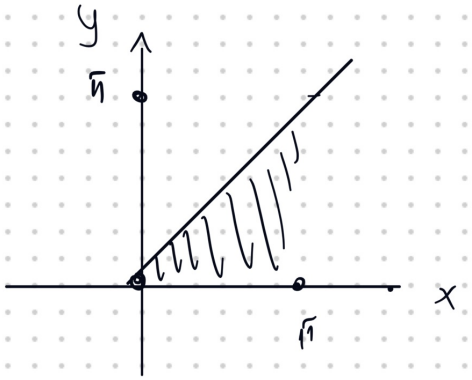
$$\iint\limits_X f(x,y)\,dx\,dy = \int\limits_a^b dx \int\limits_{\varphi(x)}^{\psi(x)} f(x,y)\,dy.$$



§8 №83.6.

$$\int\limits_{-1}^{0} dy \int\limits_{\sqrt{1-y^2}}^{2\sqrt{y+1}} f(x,y)\,dx = \int\limits_0^2 dx \int\limits_{\frac{x^2}{4}-1}^{\psi} f(x,y)\,dy$$

$$\psi(x) = \begin{cases} \sqrt{1-x^2}, & x \in [0,1] \\ 0, & x \in (1;2]. \end{cases}$$



85.2.

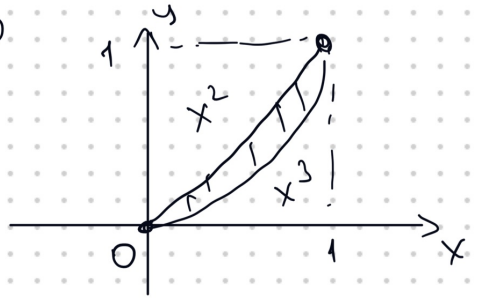
$$\int\limits_0^{\pi} dy \int\limits_y^{\pi} \frac{\sin x}{x}\,dx = \int\limits_0^{\pi} dx \int\limits_0^{x} \frac{\sin x}{x}\,dy = \int\limits_0^{\pi} dx\, \frac{\sin x}{x}\, y\Big|_0^x = \int\limits_0^{\pi} \sin x\,dx = 2.$$

10.10

№90.2.

$$\iint\limits_G \frac{y}{x^2}\,dx\,dy = \int\limits_0^1 dx \int\limits_{x^3}^{x^2} \frac{y}{x^2}\,dy = \int\limits_0^1 dx \left(\frac{x^2}{2} - \frac{x^4}{2}\right) = \frac{1}{6} - \frac{1}{10} = \frac{1}{15}$$

$G = \{0 < x,\ x^3 \le y \le x^2\}$



№90.6.

$$\iint\limits_G (x+2y)\,dx\,dy = \int\limits_2^3 dx \int (x+2y)\,dy = \int\limits_2^3 dx\, (xy+y^2)\Big|_x^{2x} =$$

$$= \int\limits_2^3 dx\,(x\cdot 2x + 4x^2 - x\cdot x - x^2) = \int\limits_2^3 4x^2\,dx = 4\,\frac{x^3}{3}\Big|_2^3 = \frac{76}{3}$$

G - обл $y = x \quad y = 2x$
$x = 2 \quad x = 3.$

10.10.

Замена переменных в кратном интеграле

Теорема

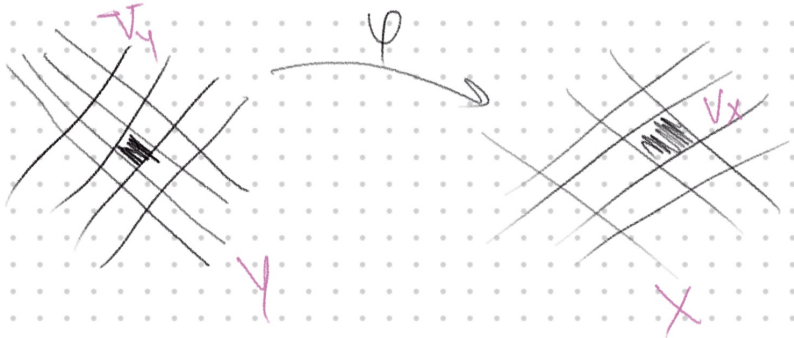
Пусть $X \subset \mathbb{R}^n$, $Y \subset \mathbb{R}^n$ измеримое, φ - отображение, переводящее Y в X, причём:

1. φ - взаимно однозначно.
2. φ - непр. диф.

Если $f(x)$ интегр. на X, тогда

$$\int_X f(x)dx = \int_Y f(\varphi(y))|\det J|\,dy.$$

$$J = \frac{\partial \varphi}{\partial x} = \begin{pmatrix} \frac{\partial \varphi_1}{\partial x_1} & \cdots & \frac{\partial \varphi_1}{\partial x_n} \\ \vdots & & \vdots \\ \frac{\partial \varphi_n}{\partial x_1} & \cdots & \frac{\partial \varphi_n}{\partial x_n} \end{pmatrix}.$$



$$|\det J| \approx \frac{V_x}{V_y}$$

№ 91.6

$$\iint_G (2y - x)\,dx\,dy = I \qquad G = \{\underbrace{y(y-x)}_{u} \le 2,\ \underbrace{x(x+y)}_{v} \le 3\}$$

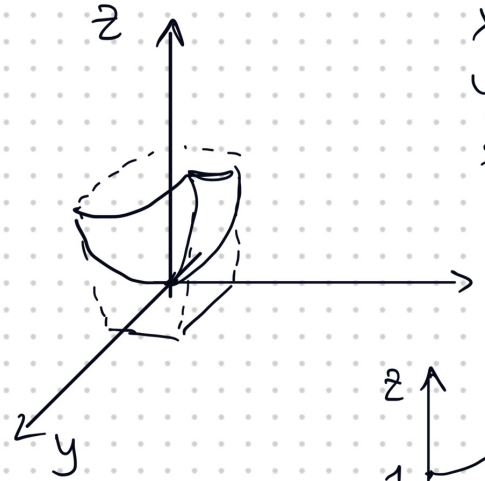
$x = u$
$y = v$

$$\det J = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1$$

$I = \iint\limits_{\widetilde{G}} (u-2v)\,|J|\,du\,dv$ $\quad \widetilde{G} = \{v(v-u)\le 2;\ u(u+v)\le 3\}$

$I = \iint\limits_{\widetilde{G}} (u-2v)\,du\,dv = -\iint\limits_{\widetilde{G}} (2v-u)\,du\,dv = -I \quad I=0$

$$\int_0^1 dx \int_0^1 dy \int_0^{x^2+y^2} f(x,y,z)\,dz$$



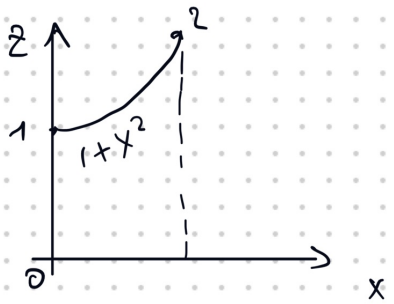
$x \in [0;1]$
$y \in [0;1]$
$z \in [0; x^2+y^2]$

а) (x,z,y)

$z \le x^2+y^2$

$y \ge \sqrt{z+x^2}$.

$z \le x^2+1$.

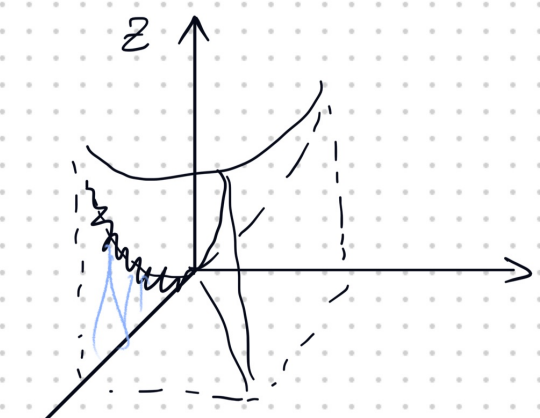


$$\int_0^1 dx \int_0^{1+x^2} dz \int_0^{\psi(x,z)} f(x,y,z)\,dy$$

$\psi(x,z)$

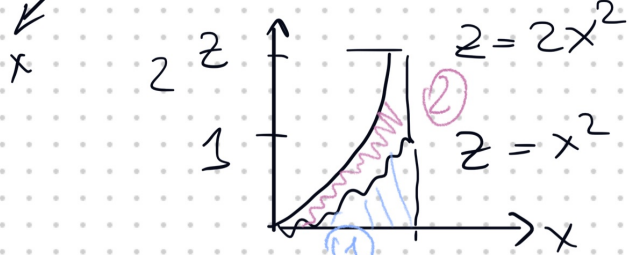
№13.3(5)

$$\int_0^1 dx \int_0^1 dy \int_0^{x^2+y^2} f\,dz \underset{(x,z,y)}{=}$$



$$= \left[\int_0^1 dx \int_0^{x^2} dz \int_0^1 f\,dy + \right.$$

$$\left. + \int_0^1 dx \int_0^{2x^2} dz \int_{\sqrt{z-x^2}}^1 f\,dy \right]$$



№ 139.3.

$$\iiint_G (x+z)\,dx\,dy\,dz$$

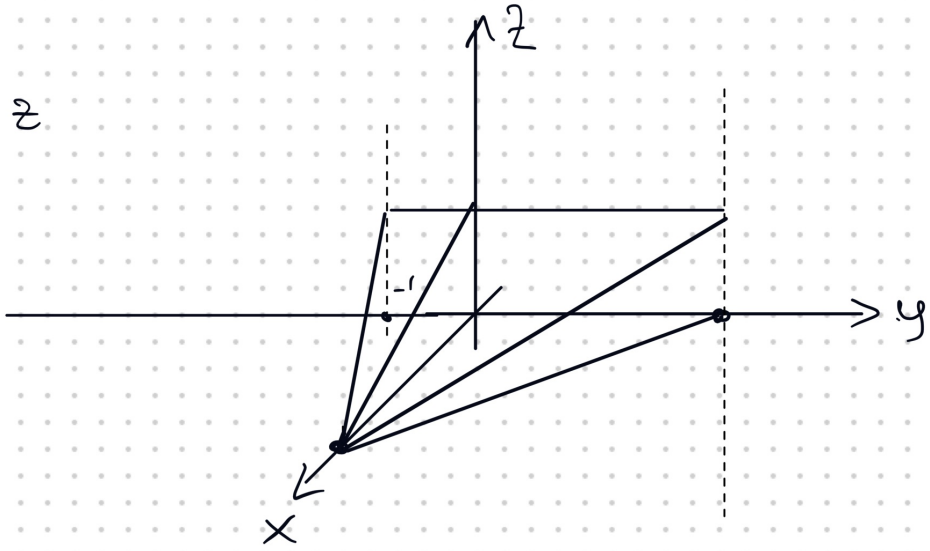
$x+y=1$

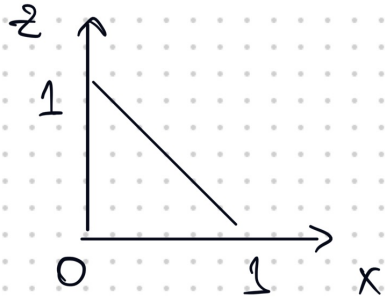
$x-y=1$

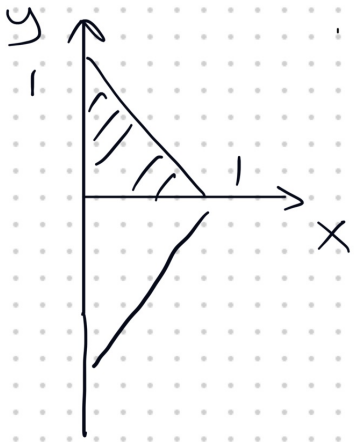
$x+z=1$

$z=0$

$x=0$.







$$= 2\int_0^1 dx \int_0^{1-x} dz \int_0^{1-x} (x+z)\,dy = 2\int_0^1 dx \int_0^{1-x} (x+z)(1-x)\,dz =$$

$$= 2\int_0^1 dx \int_0^{1-x} x(1-x) + z(1-x)\,dz = 2\int_0^1 dx \left(x(1-x)^2 + \frac{1}{2}(1-x)^3 \right) =$$

$$= 2\int_0^1 (1-x)^2 \left(x + \frac{1}{2} - \frac{x}{2} \right) dx = \int_0^1 (1-x)^2 (x+1)\,dx =$$

$$= \int_0^1 (1-x)(1-x^2)\,dx = 5/12$$

$$\int_0^a dx_1 \int_0^{x_1} dx_2 \cdots \int_0^{x_{n-1}} f(x_n)\,dx_n = \frac{1}{(n-1)!} \int_0^a f(t)(a-t)^n\,dt$$

f - непр $[0; +\infty)$.

По индукции:

$\int_0^a f(t)dt - \frac{1}{0!}\int_0^a f(t)(a-t)^{1-1}dt$ – верно.

Предположение

$$\int_0^a dx_1 \dots \int_0^{x_{k-1}} f(x_k)dx_k = \frac{1}{(k-1)!}\int_0^a f(t)(a-t)^{k-1}dt$$

$$\int_0^a dx_1 \dots \int_0^{x_{k-1}} dx_k \underbrace{\int_0^{x_k} f(x_{k+1})dx_{k+1}}_{F(x_k)}$$

$$\int_0^a dx_1 \dots \int_0^{x_{k-1}} F(x_k)dx_k = \frac{1}{(k-1)!}\int_0^a F(t)(a-t)^{k-1}dt =$$

$$= \frac{1}{(k-1)!}\int_0^a \underbrace{\int_0^t f(y)dy}_{F(t)}(a-t)^{k-1}dt = \frac{1}{(k-1)!}\int_0^a\int_0^t f(y)dy\,(a-t)^{k-1}dt =$$

$$= \frac{1}{(k-1)!}\left[-\frac{1}{k}(a-t)^k\int_0^t f(y)dy\Big|_0^a + \frac{1}{k}\int_0^a f(t)(a-t)^k dt\right] =$$

$$= \frac{1}{k!}\int_0^a f(t)(a-t)^k dt.$$

§8 №100(4)

$$\iint_G f(x,y)dxdy =$$

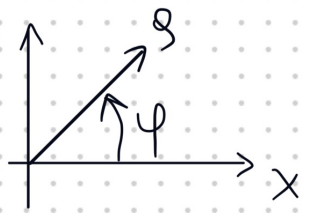
$G: x^2+y^2 \le 2ay,\ a>0.$

$dxdy = |\det J|\,d\rho\,d\varphi$

Полярные координаты:

$x = \rho\cos\varphi$

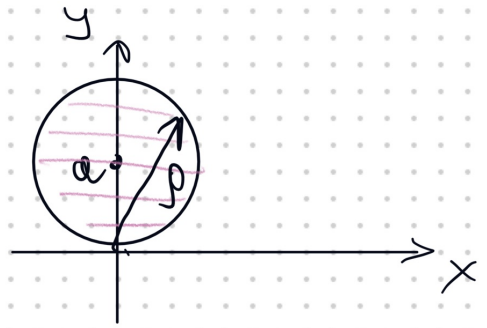
$y = \rho\sin\varphi$

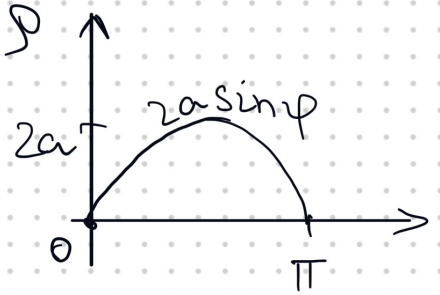


$$J = \begin{vmatrix} \frac{\partial x}{\partial \rho} & \frac{\partial x}{\partial \varphi} \\ \frac{\partial y}{\partial \rho} & \frac{\partial y}{\partial \varphi} \end{vmatrix} = \begin{vmatrix} \cos\varphi & -\rho\sin\varphi \\ \sin\varphi & \rho\cos\varphi \end{vmatrix} = \rho$$

$\Rightarrow x^2+y^2-2a\rho+a^2 \le a^2$

$x^2 + (\rho-a)^2 \le a^2.$



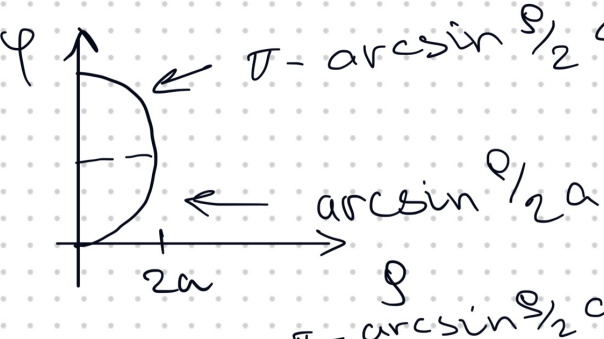


$x^2+y^2 = 2a\rho$

$\varphi = 2a\rho\sin\varphi$

$\rho = 2a\sin\varphi.$

$$= \int_0^{\pi} d\varphi \int_0^{2a\sin\varphi} f(\rho\cos\varphi, \rho\sin\varphi)\,\rho\,d\rho. =$$



$$= \int_0^{2a} \rho\,d\rho \int_{\arcsin\rho/2a}^{\pi-\arcsin\rho/2a} f(\rho\cos\varphi;\ \rho\sin\varphi)\,d\varphi$$

№107.2.

Вычислить

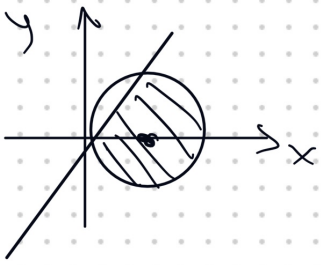
$$\iint_G y\,dx\,dy, \quad G = \{x^2+y^2 \le 2x,\ x>y\}$$

$x^2 - 2x + y^2 \le 1$

$(x-1)^2 + y^2 \le 1$

$x' = x - 1$

$y' = y$



$x' = \rho\cos\varphi$

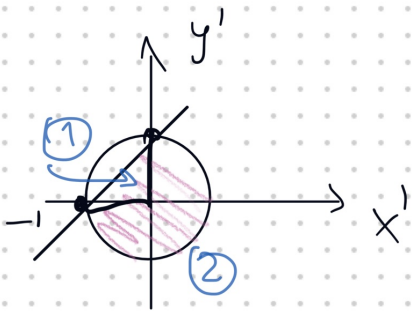
$y' = \rho\sin\varphi$

$\det J = \rho$

$$\iint y'\,dx'dy' = I_1 + I_2$$

G^*

$$G^* = \{x'^2 + y'^2 \le 1 \quad x' + 1 > y'\}$$



$$I_2 = \int_0^1 \int_{-\pi}^{\pi/2} \rho\sin\varphi\,\rho\,d\rho\,d\varphi =$$

$$= \int_0^1 \rho^2 d\rho \int_{-\pi}^{\pi/2} \sin\varphi\,d\varphi = \frac{1}{3}(-1) = -\frac{1}{3}$$

$$I_1 = \int_{-1}^{0} \int_0^{1+x} y'\,dx'dy = \int_{-1}^{0} dx' \int_0^{1+x'} y'\,dy' = \int_{-1}^{0} \frac{(y')^2}{2}\Big|_0^{1+x'} dx' =$$

$$= \int_{-1}^{0} \frac{(1+x')^2}{2} dx' = \frac{1}{6}(1+x')^3\Big|_{-1}^{0} = 1/5$$

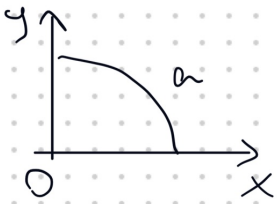
Ответ: $I = I_1 + I_2 = \frac{1}{6} - \frac{1}{3} = -1/6$

§8 №10

$$\iint_G e^{-(x^2+y^2)}\,dxdy = ; \quad G = \{x^2 + y^2 \le a\}, a > 0$$

$x \ge 0$

$y \ge 0$



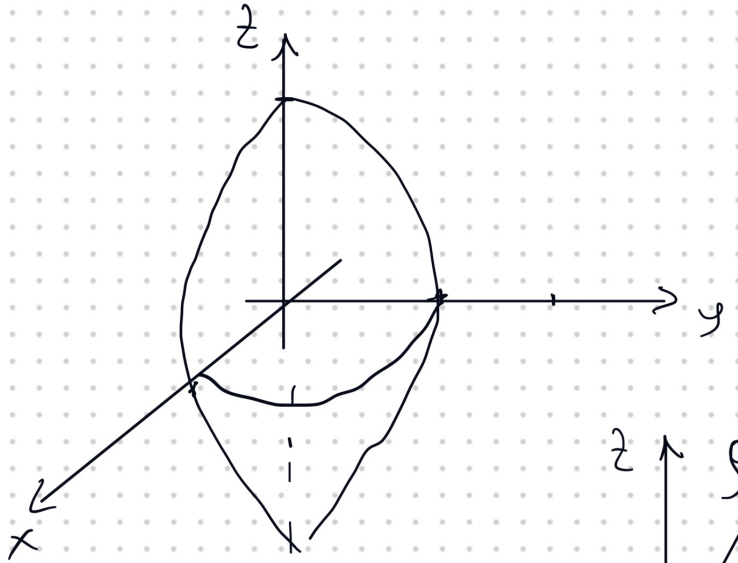
$$= \int_0^{\pi/2} d\varphi \int_0^{a} e^{-\rho^2\cos^2\varphi + \rho^2\sin^2\varphi}\,\rho\,d\rho =$$

$$= \int_0^{\pi/2} d\varphi \int_0^{a} e^{-\rho^2}\,\rho\,d\rho = \frac{\pi}{2}\cdot\frac{1}{2}e^{-\rho^2}\Big|_a^0 = \frac{\pi}{4}(1 - e^{-a^2}).$$

Доп. семинар.
22.10.

$$\iiint_G x^2+y^2-z^2\,dx\,dy\,dz = ?$$

$$G=\{1\le x^2+y^2+z^2\le 4,\ x\ge 0,\ y\ge 0\}$$

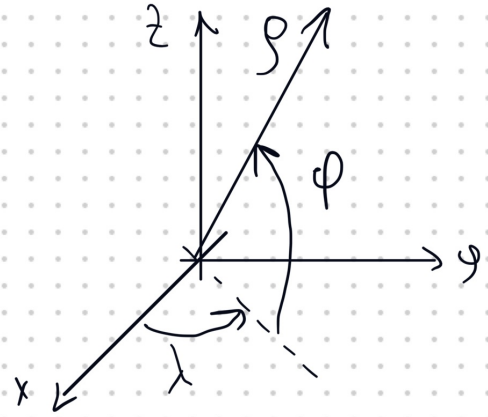


Сферични координати

$x=\rho\cos\varphi\cos\lambda$

$y=\rho\cos\varphi\sin\lambda$

$z=\rho\sin\varphi$



$$J=\begin{vmatrix} \frac{\partial x}{\partial\rho} & \frac{\partial x}{\partial\varphi} & \frac{\partial x}{\partial\lambda} \\ \frac{\partial y}{\partial\rho} & \frac{\partial y}{\partial\varphi} & \frac{\partial y}{\partial\lambda} \\ \frac{\partial z}{\partial\rho} & \frac{\partial z}{\partial\varphi} & \frac{\partial z}{\partial\lambda} \end{vmatrix} =$$

$$=\begin{pmatrix} c\varphi c\lambda & -\rho s\varphi\cos\lambda & -\rho c\varphi s\lambda \\ c\varphi s\lambda & -\rho s\varphi\sin\lambda & \rho c\varphi c\lambda \\ s\varphi & \rho c\varphi & 0 \end{pmatrix} \Rightarrow \det J=-\rho^2\cos\varphi$$

$$|\det J|=\rho^2\cos\varphi.$$

$$=\int_1^2 d\rho\int_{-\pi/2}^{\pi/2} d\varphi\int_0^{\pi/2}\rho^4(\cos^2\varphi-\sin^2\varphi)\rho^2\cos\varphi\,d\lambda =$$

$$=\int_1^2\rho^4\,d\rho\int_{-\pi/2}^{\pi/2}(\cos^2\varphi-\sin^2\varphi)\cos\varphi\,d\varphi\int_0^{\pi/2}d\lambda =$$

$$=\frac{\rho^5}{5}\Big|_1^2\int_{-\pi/2}^{\pi/2}(1-2\sin^2\varphi)\,d\sin\varphi\cdot\pi/2=\frac{31}{5}\frac{\pi}{2}\left(\sin\varphi-\frac{2}{3}\sin^3\varphi\right)\Big|_{-\pi/2}^{\pi/2}=$$

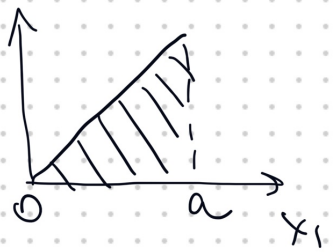
$$=\frac{31\pi}{10}\cdot\frac{2}{3}=\frac{31}{15}\pi.$$

№176.1.

$$\Pi = \{0 \le x_n \le x_{n-1} \dots \le x_2 \le x_1 \le a\}$$

$a > 0$

$n=2: 0 \le x_2 \le x_1 \le a$



$$\int_{\Pi_n} dx = \int_0^a dx_1 \int_0^{x_1} dx_2 \dots \int_0^{x_{n-1}} dx_n =$$

$$= \int_0^a dx_1 \int_0^{x_1} dx_2 \dots \int_0^{x_{n-2}} x_{n-1}\, dx_{n-1} = \int_0^a dx \int_0^{x_1} dx_2 \dots \int_0^{x_{n-2}} \frac{x_{n-2}^2}{2} dx_{n-2} =$$

$$= \dots = \frac{a^n}{n!}$$

§9 №10.

Найти S фигура

$$(a_1x + b_1y + c_1)^2 + (a_2x + b_2y + c_2)^2 \le 1.$$

$a_1b_2 - b_2a_1 \neq 0$

$$\mathcal{J} = \frac{1}{a_1b_2 - b_2a_1}$$

$\tilde{x} = a_1x + b_1y + c$

$\tilde{y} = a_2x + b_2y + c$

$$S = \iint_G dx\,dy = \iint_{\tilde{G}} |a_1b_2 - b_2a_1|^{-1} d\tilde{x}\,d\tilde{y} = \pi\,|a_1b_2 - b_2a_1|^{-1}$$

f - отобр с матрицей Якоби

f^{-1} - обратное $\mathcal{J}_{обр} = \mathcal{J}^{-1}$.

§9 №15.4.

Найти объем тела, ограничен. пов-тью:

$a, b, c > 0$

$$\left(\left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2\right)^2 + \left(\frac{z}{c}\right)^2 = 1.$$

$$V = \iiint_G f(x,y,z)\,dx\,dy\,dz$$

$x = (ar)\cos\varphi$
$y = (br)\sin\varphi$
$z = ch.$

$$J = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \varphi} & \frac{\partial x}{\partial h} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \varphi} & \frac{\partial y}{\partial h} \\ \frac{\partial z}{\partial r} & \frac{\partial z}{\partial \varphi} & \frac{\partial z}{\partial h} \end{vmatrix} =$$

$$= \begin{vmatrix} a\cos\varphi & -ar\sin\varphi & 0 \\ b\sin\varphi & br\cos\varphi & 0 \\ 0 & 0 & c \end{vmatrix} = c(abr\cos^2\varphi + abr\sin^2\varphi) =$$
$$= abcr.$$

Подставим замену:

$(r^2\cos^2\varphi + r^2\sin^2\varphi)^2 + h^2 = 1$

$r^4 + h^2 = 1 \quad r > 0$
h - любое

$r = (1-h^2)^{\frac{1}{4}}.$

$$2\int_0^1 dh \int_0^{(1-h^2)^{1/4}} dr \int_{-\pi}^{\pi} abcr\,d\varphi = 2abc\int_0^1 dh \int_0^{(1-h^2)^{1/4}} r\,dr \int_{-\pi}^{\pi} d\varphi =$$

$$= 2abc\int_0^1 dh \left(\frac{r^2}{2}\Big|_0^{(1-h^2)^{1/4}}\right)\cdot 2\pi = 2abc\pi\int_0^1 (1-h^2)^{1/2}\,dh =$$

$h = \sin\varphi$
$dh = \cos\varphi\,d\varphi$
$(1-\sin^2\varphi)^{1/2}\cos\varphi\,d\varphi$
$\cos^2\varphi\,d\varphi =$
$= \frac{1+\cos 2\varphi}{2}\,d\varphi$

$$= 2abc\pi\int_0^{\pi/2} \cos^2\alpha\,d\alpha = 2abc\pi\int_0^{\pi/2} \frac{1+\cos 2\alpha}{2}\,d\varphi =$$

$= abc\frac{\pi^2}{2}.$